# Alexandra Voroshilova

**+358 45 803 53 13** | **alex.voroshilova@gmail.com** | **Helsinki, Finland**

## PROFESSIONAL SUMMARY

Software engineer with 12+ years of experience building cloud systems, distributed services, internal platforms, and full-stack applications. Strong background in backend/API architecture, cloud infrastructure, CI/CD, observability, and production system performance. Experienced in designing and deploying reliable systems on AWS and Azure, leading cloud migrations, and coordinating cross-functional engineering work. Currently expanding into machine learning and LLM systems, with a focus on the engineering challenges of model inference, serving, and performance.

## CORE COMPETENCIES

- **Backend & Distributed Systems:** Microservices, API Architecture, Async Event Streaming, RESTful Interfaces, High-Throughput Ingestion Pipelines
- **Cloud Architecture & Infrastructure:** AWS, Microsoft Azure, Terraform (IaC), CI/CD, Infrastructure Automation, Service Deployment
- **Systems Observability & Performance:** Production Telemetry, Custom Metrics & Dashboards, Automated Alerting, Capacity Management, Latency Optimization, Performance Monitoring
- **Software Engineering:** C# (.NET/WPF), TypeScript, React, Python, SQL, Scripting
- **AI/ML:** Foundational Understanding of LLM and Deep Learning Concepts, AI-Assisted Software Development (GitHub Copilot, Cursor)

## PROFESSIONAL EXPERIENCE

**Trimble Corporation (Formerly Tekla Oy)** | *Helsinki, Finland* **Software Engineer** | *2013 – Present* **Platform Team** | *2023 – 2026*

- **Cross-Team Feature Delivery (2026):** Owned the microservice-side implementation of a new multi-model file-loading feature, managing cross-team dependencies to ensure optimal system interoperability and high serving capacity.
- **Cloud Migration (2025):** Orchestrated the technical execution and deployment design for migrating a core service from Azure to AWS with zero user downtime. Re-architected the infrastructure using Terraform to drastically improve long-term system maintainability, resulting in both lower monthly deployment costs and reduced service latency. Established robust CI/CD pipelines and built full-stack observability via custom metrics, dashboards, and automated alarms to ensure systemic reliability.

- **Standards Integration (2024):** Managed the cross-functional planning and implementation support for embedding a new BCF standard into Trimble Connect, ensuring seamless enterprise ecosystem alignment.
- **Engineering Productivity:** Integrated AI-assisted development tools to streamline code generation, optimize code reviews, and technical documentation.

**Software Engineer - Trimble Connect Web 3D Viewer & Platform Delivery** | *2019 – 2023*

- Features delivery for the core Web 3D Viewer on Trimble Connect Web using TypeScript and React. Partnered with international release teams to execute product launches, using telemetry data and performance metrics to optimize quality.

**Early Engineering & Leadership Roles (Trimble Connect)** | *2013 – 2019*

- **Product Owner & Subcontracting Coordinator (2016–2017):** Appointed as Product Owner for a critical shared 3D mobile component within a globally distributed USA-India-Finland team. Coordinated engineering requirements, influenced cross-functional stakeholders, designed APIs, and mitigated operational risks.
- **Software Engineer / Developer (2013–2016, 2017–2019):** Contributed to the core desktop, mobile, and web infrastructure of Trimble Connect and TeklaBimsight using C# and WPF. Refined code release management processes and established engineering best practices across the development team. *(Maintained part-time during M.Sc. studies from 2017–2019).*

**Kotimaa Oy** | *Helsinki, Finland* **Software Developer** | *April 2012 – 2013* Built and launched customer-facing mobile applications (Windows Phone) for a premier news site (Kotimaa24) and geographic discovery tool (Tiekirkot). Built and launched customer-facing mobile applications using JavaScript frameworks (Backbone.js, Python) to deliver scalable, responsive frontend interfaces.

## EDUCATION

**Master of Science (M.Sc.) in ICT Innovation** | *2017 – 2019* **Institutions:** KTH Royal Institute of Technology (Sweden) & Technische Universität Berlin (Germany) — *Double Degree Program.* **Specialization:** Cloud Computing & Data-Intensive Computations. **Technical Focus:** Spark, Docker, Scala, Python, AWS. **Master's Thesis:** "Comparison study on graph sampling algorithms for interactive visualizations of large-scale networks"

**Bachelor of Business Administration (BBA) in Information Technology** | *2010 – 2013* **Institutions:** Haaga-Helia University of Applied Studies (in collaboration with Aalto University)

**Moscow Aviation Institute (National Research University)** | *2009 – 2010* Undergraduate Coursework | Applied Mathematics & Physics Program

**LinkedIn:** https://www.linkedin.com/in/alexandra-voroshilova-19922747/
**Github:** https://github.com/alexvoro?tab=repositories